DARIO SHARIATIAN

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Paris, France

PhD researcher in generative modeling, working on continuous and discrete diffusion methodology. First/Co-author work at ICLR, ICML, and NeurIPS; recent Sakana AI intern with experience in multi-node H200 experiments.

EDUCATION

PhD in Computer Science, ENS Paris - PSL (INRIA), Paris — Oct 2023 – 2026
- Advised by Alain Durmus and Umut Simsekli in the Sierra team led by Francis Bach.

MSc in Mathematics - Part C, University of Oxford — 2022 – 2023

BSc/MSc in Applied Mathematics (Ingénieur Polytechnicien), École Polytechnique — 2019 – 2022

Classes préparatoires, MPSI – MP*, Lycée Saint-Louis, Paris — 2017 – 2019

RESEARCH EXPERIENCE

Research Intern at Sakana AI, with Dr. Stefano Peluchetti, Tokyo — May – Oct 2025
- Developed and first-authored Latent-Augmented Discrete Diffusion Models, adding a learnable continuous latent channel to masked discrete diffusion for up to 2x faster categorical generation.
- Designed and ran multi-node H200 experiments for large-scale text/molecule-like categorical generation.
- Owned the research loop: method design, implementation, large-scale training, ablations, paper writing.
- Proposed and co-organized the first Sakana AI research retreat, a 5-day offsite with the research staff.

Research Visit at Universita di Padova, with Prof. Giovanni Conforti — Mar 2025
- Initiated a joint mathematics/physics project on diffusion-based generative models for cosmology, spanning problem formulation, collaboration setup, and early technical direction.

Quantitative Research Intern at Squarepoint Capital, with Dr. Asgeir Birkisson, London — Mar – Aug 2022
- Developed predictive statistical models for mid-frequency equities; proposed a new machine learning approach shared across several teams.

Software Engineer Intern at Ledger, Paris — Jun – Sep 2021
- Built a C emulator for the Ledger Nano X, accelerating debugging and internal development workflows.

PUBLICATIONS

Latent-Augmented Discrete Diffusion Models (LADD) — May 2026
Shariatian, D., Durmus, A., Simsekli, U., & Peluchetti, S.
SPIGM Workshop @ ICML 2026. We introduce a learnable auxiliary latent channel for discrete diffusion, enabling cross-token modeling and improving few-step generation.

Uniform Diffusion Models Revisited — May 2026
Gourevitch, S., Janati, Y., **Shariatian, D.**, Simsekli, U., Moulines, É., Xing, E.P., & Durmus, A.
SPIGM Workshop @ ICML 2026. Leave-one-out denoising and absorbing-state reformulation for uniform diffusion, with parameterization conversions and improved sampling behavior, e.g., in language modeling.

Algorithm/Data-Dependent Generalization Bounds for Score-Based Models — Sep 2025
Shariatian, D.*, Dupuis, B.*, Haddouche, M.*, Durmus, A., & Simsekli, U.
NeurIPS 2025. Theory/empirics work establishing generalization bounds that account jointly for optimization dynamics and data distribution.

Bit-Level Discrete Diffusion with Markov Probabilistic Models — Feb 2025
Shariatian, D.*, Pham, L.T.N.*, Ocello, A., Conforti, G., & Durmus, A.
ICML 2025. Discrete diffusion for bit data, outperforming MD4 and Discrete Flow Matching on binarized MNIST with about 2.5x fewer network evaluations, alongside convergence theory.

Heavy-Tailed Diffusion with Denoising Lévy Probabilistic Models — Jan 2025

Shariatian, D., Simsekli, U., & Durmus, A.
ICLR 2025. Diffusion-model framework for heavy-tailed and imbalanced data.

Piecewise Deterministic Generative Models Jul 2024
Bertazzi, A., **Shariatian, D.**, Simsekli, U., Moulines, É., & Durmus, A.
NeurIPS 2024. Generative modeling with piecewise-deterministic Markov processes, combining deterministic motion with random jumps for smoother dynamics.

TEACHING EXPERIENCE

Teaching Assistant for Numerical Analysis, École Polytechnique Mar – Jun 2024
- Teaching assistant for MAA106 Numerical Analysis.

Oral Examiner for MSc Data Science for Business/Finance, X-HEC 2024, 2025, 2026
- Oral examiner for machine-learning and data-science students in the joint École Polytechnique / HEC program.

ACADEMIC ACHIEVEMENTS & OTHERS

MSc Mathematics with Distinction, University of Oxford 2023
BSc/MSc Applied Mathematics, Top 10%, École Polytechnique 2019-2022
- Ranked top of class in MAP553 Foundations of Machine Learning, MAP556 Monte Carlo Methods, MAP586 Modern C++ software design, and PHY430 Advanced Quantum Physics.

Classes Préparatoire MPSI / MP*, Lycée Saint-Louis 2017-2019
- Ranked 1/44 in first year and 2/42 in second year.

French National Mathematics Olympiad 2017
- Obtained a distinction in the Paris academy.

French International Mathematical Olympiad (IMO) training program, Animaths 2016
- Selected for the final French IMO team training camp at Université de Montpellier; 40 students nationwide.

TALKS, SCHOOLS, AND SERVICE

Selected Talks 2024 – 2025
- NeurIPS@Paris (oral); Sakana AI; ENS Lyon; GDR IASIS / ENS Lyon; Califrais; Alan Turing Institute; Oberwolfach MFO.

Selected Workshops and Schools 2024 – 2026
- Alan Turing Institute / Isaac Newton Institute diffusion school; Oberwolfach Statistical Challenges for Deep Generative Models; MBZUAI ML Winter School on Representation Learning and GenAI.

Service 2024 – 2026
- Reviewer: ICML 2024, NeurIPS 2024, AAAI 2025, TMLR, ICLR 2025, ICML 2025, ICML 2026 Golden Reviewer
- Organizer: diffusion models reading group at INRIA Paris; Sakana AI research retreat.

SELECTED EARLIER PROJECTS AND SKILLS

An Alternative to the Log-Likelihood, MSc thesis, University of Oxford Dec 2022 – Apr 2023
- Studied Sinkhorn EM as an entropic optimal transport alternative to maximum likelihood.

On-Board Computer for Nano-Satellite, IONSAT, École Polytechnique Space Center 2020 – 2021
- Led the team designing the OBC architecture in collaboration with CNES; presented at Dubai IAC 2021.

Programming: Python, C/C++, Qt, OpenGL. **ML stack:** PyTorch, PyTorch Lightning, Slurm, git, gdb. **Research strengths:** diffusion models, generative modeling, discrete generation, project ownership.